

1. Online convex optimization

1. Protocol and fixed comparator

Let $\mathcal{K} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex, and let the allowed losses be bounded convex functions $f_t : \mathcal{K} \rightarrow \mathbb{R}$. On round t :

1. choose $x_t \in \mathcal{K}$ before seeing f_t ;
2. reveal f_t and incur $f_t(x_t)$.

For the realized sequence,

$$R_T = \sum_{t=1}^T f_t(x_t) - \inf_{x \in \mathcal{K}} \sum_{t=1}^T f_t(x).$$

The comparator is one point fixed across all rounds and is evaluated on the same revealed losses. Write a minimum or “best fixed point” only when the infimum is attained. An adaptive loss chosen after x_t is permitted by the declared protocol; choosing x_t after seeing f_t is not.

2. Convex loss linearization

If f_t is differentiable and convex, then for every feasible comparator x ,

$$f_t(x_t) - f_t(x) \leq \nabla f_t(x_t)(x_t - x).$$

This converts a nonlinear loss gap into a gradient inner product. It is an upper bound licensed by differentiable convexity; it does not say that the current gradient points toward the hindsight comparator.

3. Euclidean projection

For nonempty closed convex $\mathcal{K}$,

$$\Pi_{\mathcal{K}}(y) = \arg \min_{x \in \mathcal{K}} \|x - y\|_2.$$

The closest point is unique, and for every feasible z ,

$$\|\Pi_{\mathcal{K}}(y) - z\|_2 \leq \|y - z\|_2.$$

Feasibility of z and convexity of the set are essential. Projection makes an outside proposal feasible, but the theorem does not give a universal running time: cost depends on how $\mathcal{K}$ is represented.

2. Projected online gradient descent

1. Update and one-step inequality

After observing the current differentiable convex loss, set $g_t = \nabla f_t(x_t)$ and update

$$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t g_t), \quad \eta_t > 0.$$

For a constant $\eta > 0$ and any $x \in \mathcal{K}$, projection and squared-distance expansion give

$$g_t^T(x_t - x) \leq \frac{\|x_t - x\|_2^2 - \|x_{t+1} - x\|_2^2}{2\eta} + \frac{\eta}{2} \|g_t\|_2^2.$$

Combine this with convex linearization before summing: regret contains loss gaps, while squared distance directly controls only gradient inner products.

2. Telescoping and the OGD theorem

Summing the one-step bound leaves only endpoints:

$$\sum_{t=1}^T (f_t(x_t) - f_t(x)) \leq \frac{\|x_1 - x\|_2^2 - \|x_{T+1} - x\|_2^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^T \|g_t\|_2^2.$$

If $T \geq 1$, $\mathcal{K}$ has diameter at most D , and $\|\nabla f_t(z)\|_2 \leq G$ for every t and feasible z , then projected OGD satisfies, for every $\eta > 0$,

$$R_T \leq \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

When $D > 0$ and $G > 0$, choose $\eta = \frac{D}{G\sqrt{T}}$ to obtain $R_T \leq DG\sqrt{T}$.

3. Parameter and computation gates

Case	Correct conclusion
$D = 0$	$\mathcal{K}$ is a singleton, hence $R_T = 0$; do not use $\eta = 0$.
$G = 0$	Each differentiable loss is constant on convex $\mathcal{K}$, hence $R_T = 0$; do not divide by zero.
$D, G > 0$	The balanced positive step is legal.

No explicit d in $DG\sqrt{T}$ does not mean dimension-free: D and G may scale with d . The regret proof also does not determine projection cost.

3. From unstable leaders to FTRL

1. FTL and its instability

Follow the Leader chooses only from past revealed losses:

$$x_t \in \arg \min_{x \in \mathcal{K}} \sum_{s=1}^{t-1} f_s(x).$$

A tie-breaking rule is part of the implementation. Including f_t in this objective before playing x_t violates the online information order.

FTL can have linear regret. On $\mathcal{K} = [-1, 1]$, take $f_1(x) = \frac{x}{2}$ and, for $t \geq 2$, alternate $-x$ on even rounds and x on odd rounds. With the stated round-1 tie choice $x_1 = 1$, the past leader alternates endpoints and has exact regret T . Changing only the first tie cannot remove the later jumps.

2. Strong convexity is curvature

Relative to a declared norm $\|\cdot\|$, differentiable $\mathcal{R}$ is α -strongly convex, $\alpha > 0$, when

$$\mathcal{R}(y) \geq \mathcal{R}(x) + \nabla \mathcal{R}(x)^T(y - x) + \frac{\alpha}{2} \|y - x\|^2.$$

For nonsmooth convex $\mathcal{R}$, use a valid subgradient at x . Strong convexity supplies a quadratic lower gap; it is not smoothness, differentiability, or uniqueness of a gradient. On $[-1, 1]$, $\mathcal{R}(x) = \frac{x^2}{2}$ is 1-strongly convex in the Euclidean norm, whereas $|x|$ is convex and nonsmooth but not strongly convex.

3. Linearized FTRL update

Given $\eta > 0$ and a strongly convex regularizer $\mathcal{R}$,

$$x_t = \arg \min_{x \in \mathcal{K}} \left[\eta \sum_{s=1}^{t-1} g_s^T x + \mathcal{R}(x) \right], \quad g_s = \nabla f_s(x_s).$$

Only past linearized gradients may enter x_t . The regularizer makes the leader resist small changes in cumulative loss. In the signed example with $\mathcal{R}(x) = \frac{x^2}{2}$ and $0 < \eta \leq 2$, $x_t = \text{clip}(-\eta \sum_{s < t} g_s, -1, 1)$: later decisions alternate between $-\frac{\eta}{2}$ and $\frac{\eta}{2}$, rather than between endpoints. Stability alone does not license logarithmic regret.

4. FTRL regret tradeoff

Let x_1 minimize $\mathcal{R}$ and $\Delta_u = \mathcal{R}(u) - \mathcal{R}(x_1) \geq 0$. Under the source's bounded closed convex set and strongly convex, smooth, twice-differentiable regularizer setup, the registered FTRL update obeys

$$R_{T(u)} \leq 2\eta \sum_{t=1}^T \|g_t\|_t^* 2 + \frac{\Delta_u}{\eta}.$$

Here $\|\cdot\|_t$ is the local norm induced by the regularizer curvature and $\|\cdot\|_t^*$ is its dual; these are not automatically the global Euclidean norm. The two terms are respectively the movement bill and comparator regularizer geometry.

If $\|g_t\|_t^* \leq G_{\mathcal{R}}$ and $\Delta_u \leq \Delta_{\mathcal{R}}$ uniformly, then

$$R_{T(u)} \leq 2\eta G_{\mathcal{R}}^2 T + \frac{\Delta_{\mathcal{R}}}{\eta}.$$

For positive $\Delta_{\mathcal{R}}, G_{\mathcal{R}}$, balancing with $\eta = \sqrt{\frac{\Delta_{\mathcal{R}}}{2G_{\mathcal{R}}^2 T}}$ gives order $\sqrt{T}$.

5. Why the movement bound works

For the regularizer's first-order remainder, write

$$B_{\mathcal{R}}(x, y) = \mathcal{R}(x) - \mathcal{R}(y) - \nabla \mathcal{R}(y)^T (x - y).$$

The source's local curvature coordinates satisfy $B_{\mathcal{R}}(x_t, x_{t+1}) = \frac{\|x_t - x_{t+1}\|_t^2}{2}$. Consecutive-leader optimality gives

$$B_{\mathcal{R}}(x_t, x_{t+1}) \leq \eta g_t^T (x_t - x_{t+1}).$$

Local dual Cauchy–Schwarz yields $g_t^T (x_t - x_{t+1}) \leq 2\eta \|g_t\|_t^* 2$.

Substituting this into

$$R_{T(u)} \leq \sum_t g_t^T (x_t - x_{t+1}) + \frac{\Delta_u}{\eta}$$

shows the two jobs: the regularizer controls successive movement, while the comparator pays through $\frac{\Delta_u}{\eta}$.

4. Which rates are licensed?

1. Match structure, schedule, and method

Loss structure	Licensed mechanism	Rate
general convex	fixed strong $\mathcal{R}$ + balanced FTRL	$O(\sqrt{T})$
α -strongly convex	projected OGD, $\eta_t = \frac{1}{\alpha t}$	$R_T \leq \frac{G^2}{2\alpha} (1 + \log T)$
α -exp-concave	$\exp(-\alpha f_t)$ concave + matching EWOO/second-order method	$O(\log T)$

For strongly convex losses, the negative term $-\alpha \frac{\|x_t - u\|_2^2}{2}$ cancels successive distance coefficients under the matching schedule, leaving $\sum_t \frac{1}{t} \leq 1 + \log T$. A strong regularizer alone, a decreasing step without loss curvature, or exp-concavity paired with an arbitrary first-order update does not prove a logarithmic rate.

2. General square-root lower bound

For every OCO algorithm there exists a bounded convex decision problem with bounded linear gradients whose worst-case regret is

$$\Omega(DG\sqrt{T}).$$

One stationary construction uses $\mathcal{K} = \{x \in \mathbb{R}^n : \|x\|_{\infty} \leq 1\}$ and independent uniform sign vectors $v_t \in \{-1, 1\}^n$ with $f_{t(x)} = v_t^T x$. The learner's expected loss is zero because v_t is fresh after x_t is fixed. The hindsight comparator loses $-\sum_i \left| \sum_t v_{t(i)} \right|$, whose expectation has order $-n\sqrt{T}$. Since $D \leq 2\sqrt{n}$ and $G \leq \sqrt{n}$, this matches the $DG\sqrt{T}$ scale up to constants.

The result applies to unrestricted bounded OCO; it does not contradict faster rates under additional curvature or exp-concavity assumptions.

5. From online regret to population risk

1. A valid online-to-batch bridge

Sequence-wise regret and population risk are different objects. A valid bridge assumes:

1. iid examples z_t from one distribution;
2. a convex decision/hypothesis class and bounded loss $\ell \in [0, 1]$ convex in the decision;
3. h_t is chosen before z_t , and $\bar{h} = T^{-1} \sum_t h_t$;
4. concentration for the adaptive iterates and fixed comparator.

Under the source's reduction, with probability at least $1 - \delta$,

$$R(\bar{h}) \leq R(h^*) + \frac{\text{Regret}_T}{T} + \sqrt{\frac{8 \log(\frac{2}{\delta})}{T}}.$$

Convexity licenses averaging; iid sampling and boundedness support the risk and concentration steps. Sublinear regret alone is not a population generalization theorem, and $\frac{R_T}{T}$ is not the same object as $R(h)$.

2. Keep algorithm identities separate

Method	Update object	Do not identify with
OGD	gradient step + Euclidean projection	general FTRL without matching geometry
FTRL	past linearized gradients + $\mathcal{R}$	FTL on original nonlinear losses
Hedge	exponential expert weights	linear MWU by name alone
linear MWU	$w_{t+1}(i) = w_{t(i)}(1 - \eta \ell_{t(i)})$	Hedge or AdaBoost by name alone
AdaBoost	training-example distribution + weak learner	an OCO method with identical iterates

Shared potential reasoning or comparator language does not make update states, losses, or iterates identical.